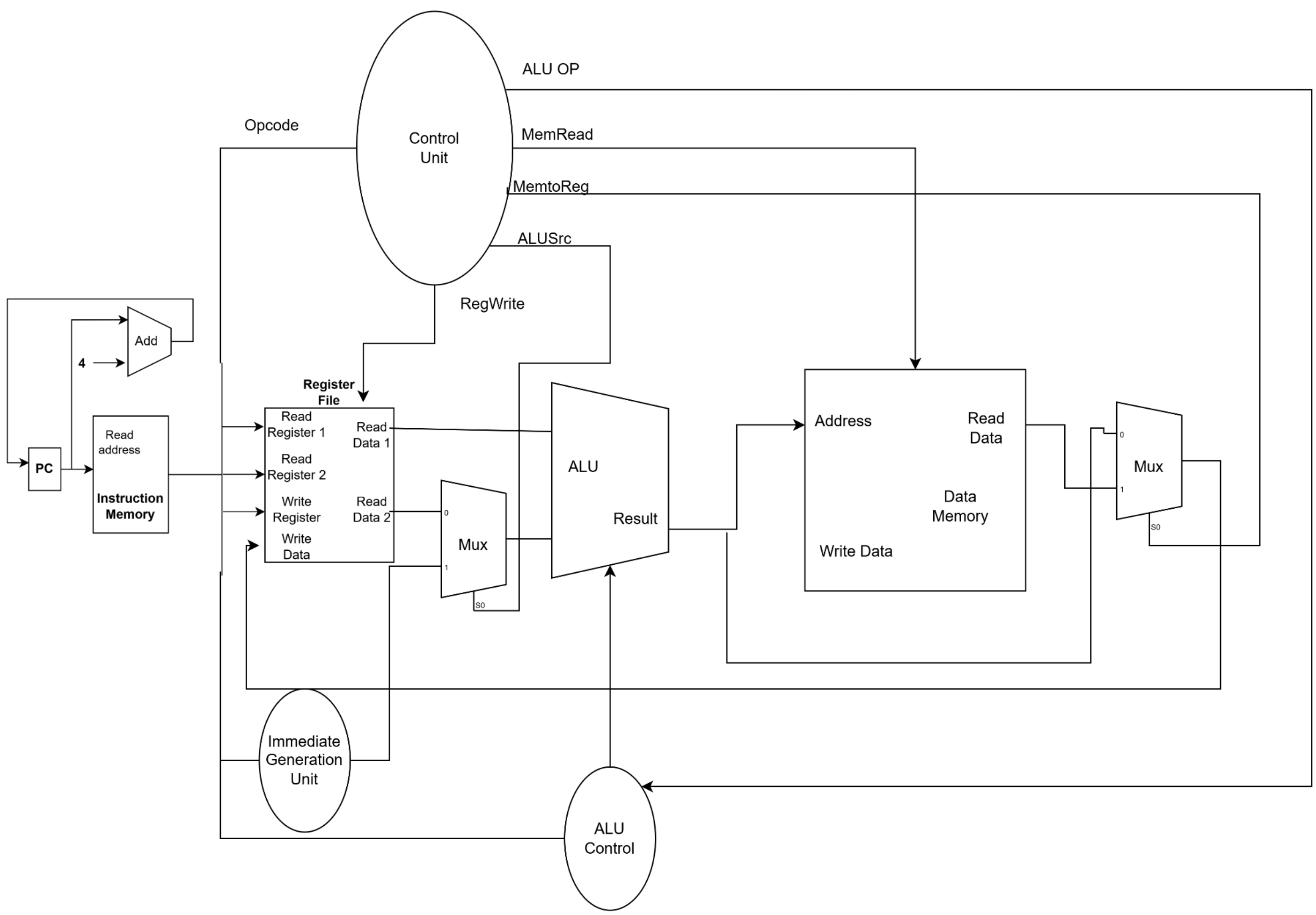
CSE340: Computer Architecture
Spring 2026
Quiz - 03
Duration: 30 minutes

A

Name:	ID:	Section:

1. Draw and complete the Datapath by adding missing components and wiring so that it works for both **R-type Instructions and Load instruction**. You must label the components, wires and mention the control signals. [8]



2. Write the correct control signals for the following set of instructions when executed in a single cycle datapath.

Instruction	RegWrite	MemRead	MemWrite	Branch	MemtoReg	ALU OP	
and x5,x6,x7	1	0	0	0	0	10	[1]
lw x18, 10(x20)	1	1	0	0	1	00	[2]
Beq x20, x21, L1	0	0	0	1	x	01	[1]

3. Consider the following instruction mix given below. In a single cycle Datapath scenario, answer the following questions: [3]

R-type	I-type (non-load)	Load	Store	Branch
15%	25%	15%	15%	30%

i. For **what** fraction of all instruction register write signals are enabled?
ii. For **what** fraction of instructions does the final output come directly from the ALU?
iii. For **what** fraction of all instruction data needs to perform immediate sign extension?

i. Register write = r-type + I-type +load = 15+25+15 = 55%
ii. ALU final output = r-type + I-type = 15+25 = 40 %
iii. Sign extension = I-type + load+store+branch = 25+15+15+30 =85%

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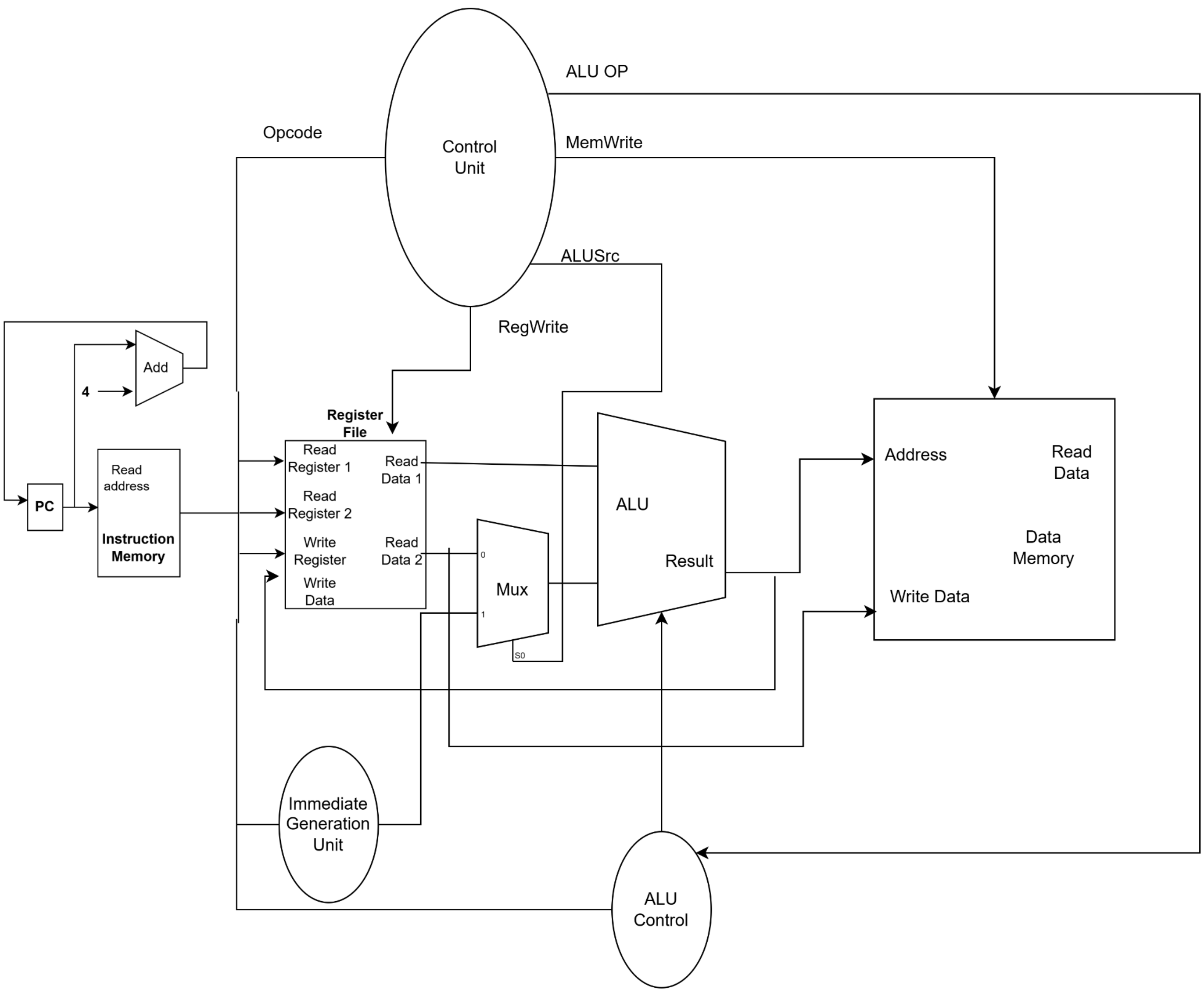
B

Name:	ID:	Section:

1. Draw and complete the Datapath by adding missing components and wiring so that it works for both **R-type Instructions and Store instruction**. You must label the components, wires and mention the control signals.



2. Write the correct control signals for the following set of instructions when executed in a single cycle datapath.

Instruction	RegWrite	MemRead	MemWrite	Branch	MemtoReg	ALU OP	
or x8,x9,x10	1	0	0	0	0	10	[1]
lb x20, 16(x22)	1	1	0	0	1	00	[2]
Beq x17, x18, L2	0	0	0	1	x	01	[1]

3. Consider the following instruction mix given below. In a single cycle Datapath scenario, answer the following questions: [3]

R-type	I-type (non-load)	Load	Store	Branch
20%	10%	35%	20%	15%

i. For **what** fraction of all instruction register write signals are enabled?
ii. For **what** fraction of instructions does the final output come directly from the ALU?
iii. For **what** fraction of all instruction data needs to perform immediate sign extension?

i. Register write = r-type + I-type +load = 20+10+35= 65%
ii. ALU final output = r-type + I-type = 20+10 = 30 %
iii. Sign extension = I-type + load+store+branch = 10+35+20+15 =80%